Custom 4-Bit Microprocessor - ISA Reference

This document serves as the official Instruction Set Architecture (ISA) reference manual for the custom 4-bit CPU. It details the binary opcodes, data paths, and hardware execution logic for the Arithmetic Logic Unit (ALU) and Stack operations.

1. Logical Operations

Instruction	Binary Opcode	Data Inputs Used	Functional Hardware Output Description
AND	0000	TempA, TempB	Performs a bitwise logical AND operation on each bit position.
OR	0001	TempA, TempB	Performs a bitwise logical OR operation on each bit position.
XOR	0010	TempA, TempB	Performs a bitwise logical XOR (exclusive OR) operation.
NOT	0011	TempA	Inverts all bits in TempA (1s complement). TempB is ignored.

2. Arithmetic Operations

Instruction	Binary Opcode	Data Inputs Used	Functional Hardware Output Description
ADD	0100	TempA, TempB	Adds the values of TempA and TempB (A + B).
SUB	0101	TempA, TempB	Subtracts TempB from TempA (A - B) using internal 2's complement hardware logic.

3. Advanced & Bitwise Shift Operations

Instruction	Binary Opcode	Data Inputs Used	Functional Hardware Output Description
SHL	1000	TempA	Shifts the bits of TempA left by one position (A << 1).
SHR	1001	TempA	Shifts the bits of TempA right by one position (A >> 1).
INC	1010	TempA	Increments the current value of TempA by 1.
DEC	1011	TempA	Decrements the current value of TempA by 1.

4. Stack & Register Operations

Instruction	Binary Opcode	Data Inputs Used	Functional Hardware Output Description
PUSH AX	1100	AX Register	Copies the current 4-bit value from the AX register and pushes it onto the top of the Stack.
PUSH BX	1101	BX Register	Copies the current 4-bit value from the BX register and pushes it onto the top of the Stack.
POP AX	1110	Stack (Top)	Pops the top 4-bit value from the Stack and loads it directly into the AX hardware register.
POP BX	1111	Stack (Top)	Pops the top 4-bit value from the Stack and loads it directly into the BX hardware register.